

Coursework Instructions: Traversing the Maze

Tasks

Navigating through a maze while obeying traffic rules and counting objects in designated areas (Fig. 1).

1. **Starting Point:** The robot will begin at its designated starting position.
2. **Oranges Sector:** The robot will then proceed to the sector with the poster with oranges while obeying traffic rules. Upon detecting an orange, the robot should slow down to ensure accurate counting. After counting all oranges, the robot should resume its navigation through the maze, adjusting its speed accordingly.
3. **Trees Sector:** Following the "oranges sector", the robot will move to the "trees sector". Here, the camera should be used to count the number of visible trunks.
4. **Vehicles Sector:** Next, the robot will travel to the "vehicles sector". Within this area, the camera should be used to identify and count all vehicles.
5. **Ending Point:** After completing the vehicle count, the robot should proceed to the designated end section and come to a complete stop.

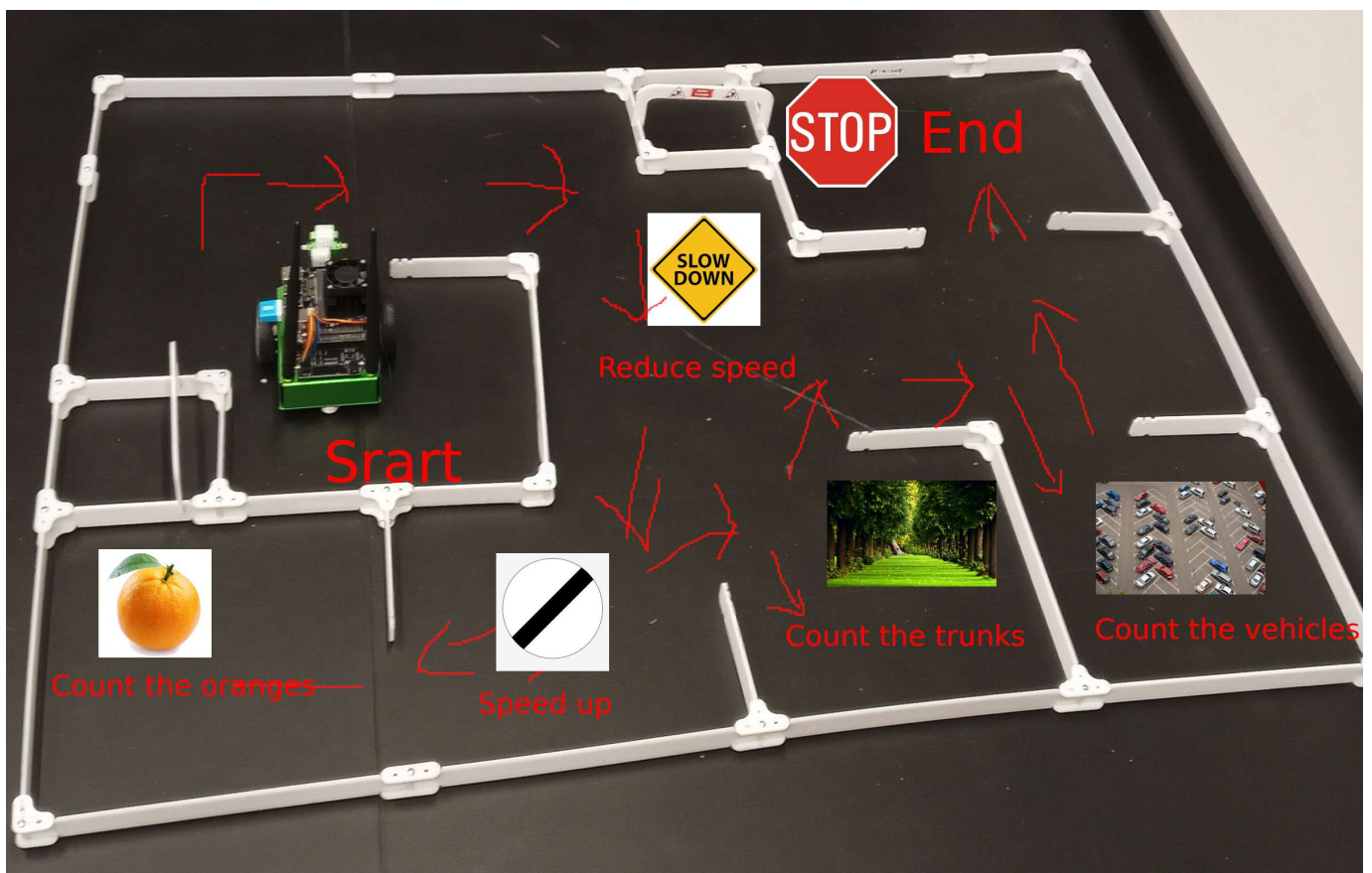


Fig. 1 - Real-world maze.

You are expected to demonstrate **as a team** the features you were able to implement from the list below with a live demo. Note that each of you will be individually evaluated on how you describe your team's approach, design, evaluation of performance, and code during the demo when prompted.

The end goal is to produce the kind of behaviours and functionalities seen in action in the video shown below (**note, however, that in your case all behaviours must be automated!**):

For this to work, **you will need to implement visual odometry (VO) -- to this end, we recommend using RTABMAP for ROS2** (https://github.com/introlab/rtabmap_ros).

References

[1] Available online, https://jetbot.org/master/examples/basic_motion.html, last accessed: 14/03/2024

[2] Available online, <https://github.com/IntelRealSense/realsense-ros/tree/ros2-legacy> , last accessed: 14/03/2024

[3] Available online, https://github.com/introlab/rtabmap_ros, last accessed: 04/03/2025

[4] Available online, RTABMAP ROS2 tutorial -- <https://www.youtube.com/watch?v=N870IyPgg3c>, last accessed: 04/03/2025

[5] Available online, RTABMAP ROS2 outdoor example -- <https://www.youtube.com/watch?v=9ZmPg6WV7KI>, last accessed: 04/03/2025